

Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a) A relator $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories: it preserves 1 , composition and $\sqsubseteq$.

	the statement	(5.1b)
1	$\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$. Lemma 5.1	
2	Over a tabular allegory a functor is a relator $\iff$ it preserves $^\circ$. Theorem 5.1 the book’s reverse direction (preserving $^\circ \implies$ relator) needs the source to be Rel : the counterexample over a tabular allegory that is not Rel shows it can fail	
3	$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.	
4	Two relators agreeing on maps are equal. Corollary 5.1	
5	$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$ Ex 5.2	
6	$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly. Ex 5.2	
7	$F(\text{dom}(R)) = \text{dom}(F(R))$ Ex 5.5	

The **power relator** $P \multimap xs \ P(R) \ ys \iff (\forall a \in xs. \ \exists b \in ys. \ a \ R \ b) \wedge (\forall b \in ys. \ \exists a \in xs. \ a \ R \ b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

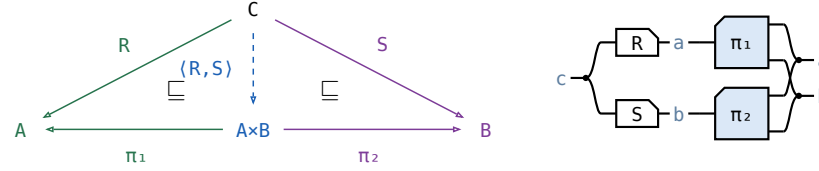
§5.2. Relational products

$$\langle R, S \rangle \triangleq (R \pi_1^\circ) \cap (S \pi_2^\circ) \text{ — the fork of } R : C \rightarrow A \text{ and } S : C \rightarrow B, \text{ with } (\pi_1, \pi_2) \text{ the tabulation of } \tau. \quad (5.1)$$

$$\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ \quad c \in \langle R, S \rangle \iff c R a \wedge c S b. \quad (5.2a)$$

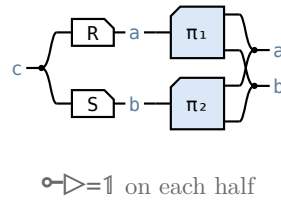
$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad (5.6)$$

$$\langle R, S \rangle \pi_2 = \text{dom}(R) S \quad (5.7)$$



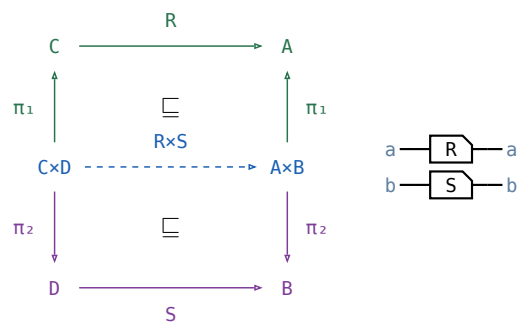
A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In $\mathbf{Rel}$, $c \in \langle R, S \rangle \iff c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft(R \otimes S)$ on the right.

No $^\circ$ survives the translation. $\pi_1 = 1 \otimes \bullet$ discards the second component, so $\pi_1^\circ = 1 \otimes \bullet$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§5.2.1. Relational product $R \times S$

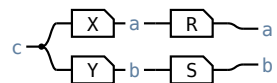
$$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle \text{ — the relational product } R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle. \quad (5.2)$$



Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) \in (R \times S) \iff c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§5.2.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:

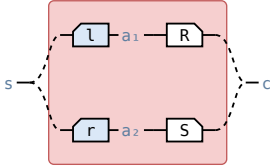
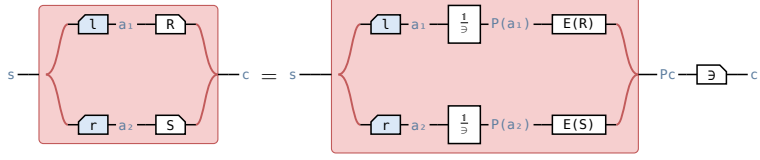
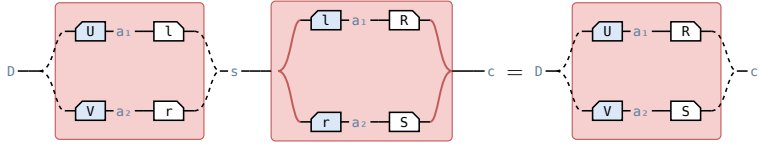


(5.2.2a)

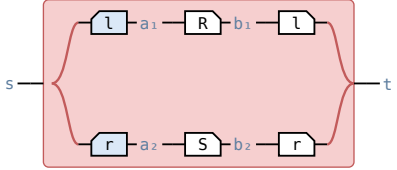
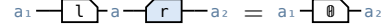
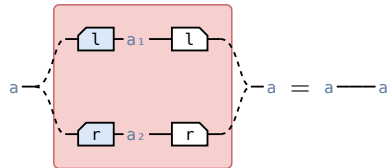
	the law	picture
1	$\langle X, Y \rangle \langle R, S \rangle^\circ = (X \ R^\circ) \cap (Y \ S^\circ)$	
2	$\langle R, S \rangle^\circ \langle U, V \rangle \sqsubseteq (R^\circ U) \times (S^\circ V)$	
3	$\text{Map}(f) \Rightarrow f \langle R, S \rangle = \langle f \ R, f \ S \rangle$ f a map; it fails for an arbitrary arrow;	
4	$X \langle R, S \rangle \sqsubseteq \langle X \ R, X \ S \rangle$ the half of the row above that holds for every X	
5	$F(R \times S) \text{ unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$; only $\sqsubseteq$ for an arbitrary R, S	
6	$g = \text{curry}(f) \iff (g \times 1) \text{ eval} = f$ reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?	

§5.3. Relational coproducts

(5.3a)

	the statement $R : a_1 \rightarrow c, S : a_2 \rightarrow c$	picture 
1	$[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$ (5.9) The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
2	$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
3	$l [R, S] = R, r [R, S] = S$ $l T = R \wedge r T = S \Rightarrow T = [R, S],$	
4	$U : a_1 \rightarrow D, V : a_2 \rightarrow D$ $[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S) \quad (5.11)$	
5	$Z : c \rightarrow D$ $[R, S] Z = [R Z, S Z]$	

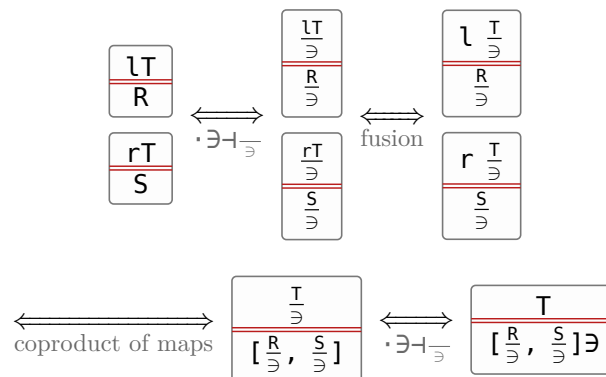
(5.3b)

	the statement $R : a_1 \rightarrow b_1, S : a_2 \rightarrow b_2$	picture 
1	$R + S \triangleq [R l, S r] \quad (5.10)$	
2	$l l^\circ = 1, r r^\circ = 1 \quad \text{Ex 5.12}$	
3	$l r^\circ = 0, r l^\circ = 0 \quad \text{Ex 5.12}$	
4	$(l^\circ l) \cup (r^\circ r) = 1 \quad \text{Ex 5.12}$	

§5.3.1. $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$ — a case of relations is the case of their transposes, then $\exists$

The universal-property row is not free: l, r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\exists$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow PC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(5.3.1a)



(5.3.1b)



Nothing here holds only up to $\cong$: every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{S}, \frac{S}{S}] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\bar{l} \circ \frac{R}{S} \cup \bar{r} \circ \frac{S}{S}) \exists = \bar{l} \circ R \cup \bar{r} \circ S$.

(5.3.1c)

the law	picture
1 prove (5.11), and say why duality does not carry it over from the product law	
2 $(R + S) \cap ([P, Q] [U, V]^\circ)$ $= (R \cap (P \cup^\circ)) + (S \cap (Q \cup^\circ))$ $[P, Q] [U, V]^\circ$ is a full 2×2 of composites; $R+S$ is diagonal, so the meet cuts the two off-diagonal branches	The diagram illustrates the law $(R + S) \cap ([P, Q] [U, V]^\circ) = (R \cap (P \cup^\circ)) + (S \cap (Q \cup^\circ))$. It shows a complex string diagram with multiple boxes and wires, representing the composition of relations and their meet operation. The diagram is divided into two main parts by an equals sign, showing the equivalence of two ways to compute the meet.

§5.4. The power relator $P(R)$

(5.4a)

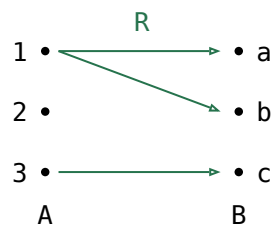
$P(R) \triangleq (\epsilon \setminus (R \epsilon)) \cap ((\exists R) / \exists)$ — the power relator $P(R) : PA \rightarrow PB$, R lifted to sets.

$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$

(5.4b)

	in Rel	in words	$\text{im}(xs)$	partner
1	$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
2	$(\exists / (\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
3	$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
4	$((\exists R^\circ) / \exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	—	every x has a partner
5	$P(R)$	$P(R)(xs, ys) \iff (\forall y. y \in ys \implies (\exists x. x \in xs \wedge R(x, y))) \wedge (\forall x. x \in xs \implies (\exists y. y \in ys \wedge R(x, y)))$	—	every x and every y has a partner

(5.4c)



	xs	$(\exists R) / \exists$	$(\exists / (\exists R))^\circ$	$E(R)$	$((\exists R^\circ) / \exists)^\circ$	$P(R)$
1	$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
2	$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
3	$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
4	$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
5	$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
6	$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
7	$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
8	$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

(5.4d)

	law	the reading
1	$X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
2	$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
3	$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
4	$P(R \ S) = P(R) \ P(S)$	$\supseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§5.4.1. Type relators

(5.4.1a)

$T(R) = \langle F(R, 1) \ \alpha \rangle$ — the type functor T acts on an arrow $R : A \rightarrow B$ by rebuilding with α , applying R to the parameter.

(5.4.1b)

name	law	what it says
1 the defining equation	$T(R) = \langle F(R, 1) \ \alpha \rangle$	Rebuild the structure with α , applying R to the parameter on the way.
2 type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

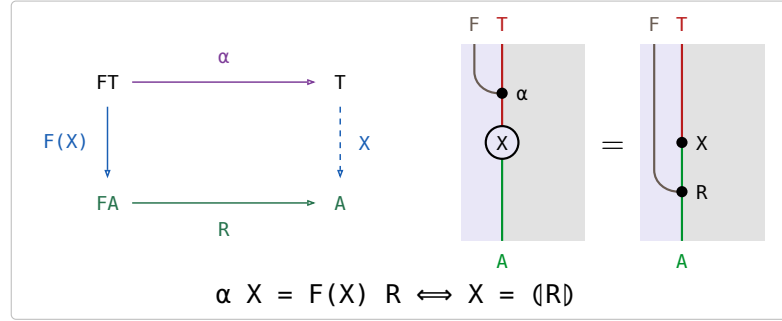
§5.5. Relational catamorphisms

(5.5a)

The initial algebra $\alpha : F(T) \rightarrow T$ among the maps is also initial in the allegory:

§5.5.1. The defining equation

(5.5.1a)



An F-algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F-Algebra only need it works on T and A .

(5.5.1b)

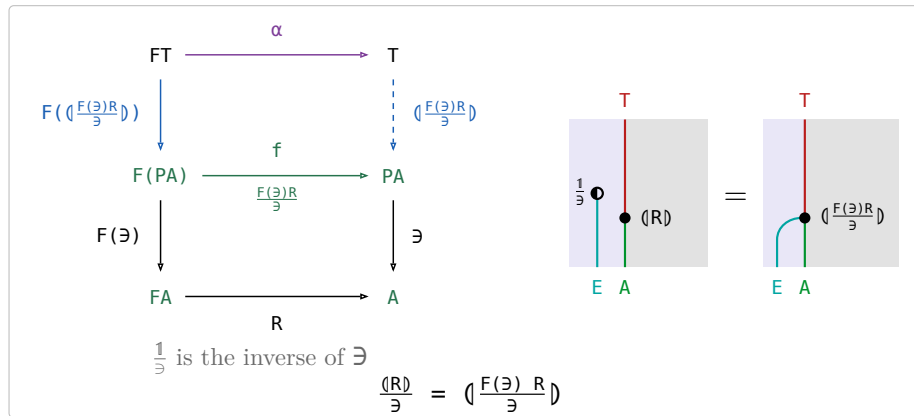
$$\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$$

catamorphism: arrows of the form $\llbracket f \rrbracket$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing

1	part	Nat	[A]
2	datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
3	base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
4	initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
5	the fold, pointwise	$\llbracket [c, f] \rrbracket (\text{zero}) = c$ $\llbracket [c, f] \rrbracket (\text{succ}(n)) = f(\llbracket [c, f] \rrbracket (n))$	$\llbracket [c, f] \rrbracket (\text{nil}) = c$ $\llbracket [c, f] \rrbracket (\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket (x))$

§5.5.2. $\llbracket R \rrbracket = \llbracket \frac{F(\exists) R}{\exists} \rrbracket \exists$ — folding with R is a fold of maps into sets, then $\exists$

(5.5.2a)



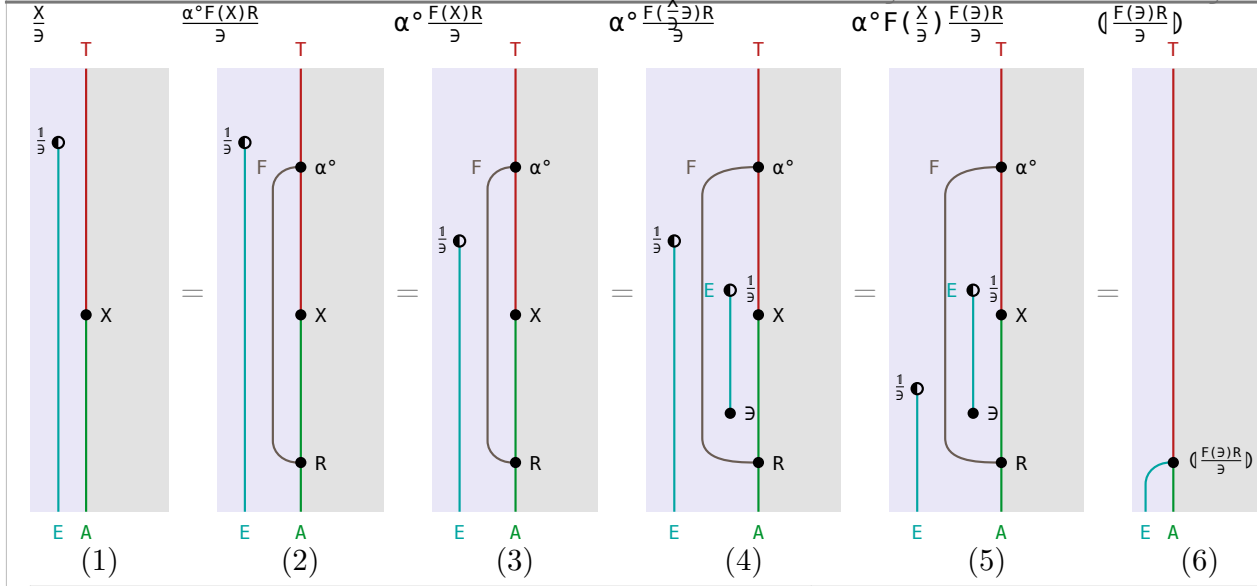
(5.12)

$$\begin{array}{c}
 \frac{\alpha X}{F(X)R} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \frac{\frac{\alpha \frac{X}{\exists}}{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}}}{\frac{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}}{\exists}} \xleftrightarrow{\text{reduce of maps}} \frac{\frac{X}{\exists}}{\frac{\llbracket \frac{F(\exists)R}{\exists} \rrbracket}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}
 \end{array}$$

the last two rows at $X := \llbracket R \rrbracket$:

$$\alpha X = F(X) R \iff \frac{X}{\exists} = \langle \frac{F(\exists) R}{\exists} \rangle \quad (5.5.2b)$$

in a tabular allegory, with F a relator and $\alpha : F(T) \rightarrow T$ its initial algebra: a relation X out of T satisfies the fold equation $\alpha X = F(X) R$ of R exactly when $\frac{X}{\exists}$ is the fold of the map $\frac{F(\exists) R}{\exists}$



1	$\frac{X}{\exists}$	α iso; $\alpha X = F(X) R$
2	$= \frac{\alpha^0 F(X) R}{\exists}$	α^0 is a map because α is an iso, so it leaves Λ : $\frac{\alpha^0 F(X) R}{\exists} = \alpha^0 \frac{F(X) R}{\exists}$ — $\text{Map}(f)$ $\implies \frac{f R}{\exists} = f \frac{R}{\exists}$
3	$= \alpha^0 \frac{F(X) R}{\exists}$	$X = \frac{X}{\exists} \frac{R}{\exists} \implies \frac{R}{\exists} \exists = R$
4	$= \alpha^0 \frac{F(\frac{X}{\exists}) R}{\exists}$	$F(\frac{X}{\exists}) = F(\frac{X}{\exists}) F(\exists)$ as F is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves Λ — $\triangleq$ relator, $\text{Map}(f)$ $\implies \frac{f R}{\exists} = f \frac{R}{\exists}$
5	$= \alpha^0 F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$	fold uniqueness, $\frac{F(\exists) R}{\exists}$ being a map — $(h : f \rightarrow g) \triangleq f h = F(h) g$, $\text{Map}(f)$ $\implies \alpha \langle f \rangle = F(\langle f \rangle) f$
6	$= \langle \frac{F(\exists) R}{\exists} \rangle$	

§5.6. Combinatorial functions

(5.6a)

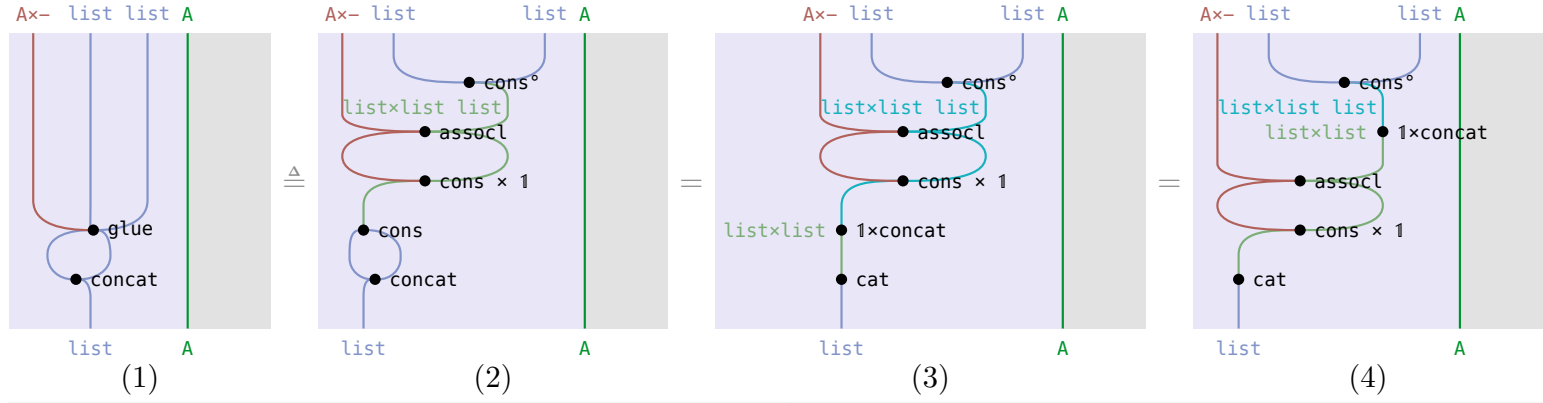
	definition	type	note
1	$[A] ::= \text{nil} \mid \text{cons}(A, [A]) \quad \text{list}(A) = [A]$	$\mathcal{A} \rightarrow \mathcal{A}$	The list type, under the short name it keeps.
2	$\text{list}(R) = \langle [\text{nil}, (R \times 1) \text{ cons}] \rangle$	$[A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.
3	$\text{subseq} = \langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$	$[A] \rightarrow [A]$	$xs \text{ subseq } ys$: ys is xs with elements dropped — cons keeps the head, π_2 drops it.
4	$\text{prefix} = \langle [\text{nil}, \text{cons} \cup \rightarrow \text{nil}] \rangle$ $\text{prefix} = \text{cat}^\circ \pi_1 = \text{init}^*$	$[A] \rightarrow [A]$	ys is an initial segment of xs ; the first nil is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$.
5	$\text{suffix} = \text{cat}^\circ \pi_2 = \text{tail}^*$	$[A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$; as a reduce it needs snoc -lists.
6	$\text{partition} = \text{concat}^\circ$	$[A] \rightarrow [[A]]$	This cat is restricted to $[A]^+ \times [A] \rightarrow [A]$, so ys is a list of non-empty segments of xs .
7	$\text{concat} = \langle [\text{nil}, \text{cat}] \rangle$	$[[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.
8	inits	$[A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\exists}$, listing the prefixes by increasing length.
9	tails	$[A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.
10	$\text{cpp}, \text{cpl}, \text{cpr}$	$[A] \times [B] \rightarrow [A \times B]$ $[A] \times B \rightarrow [A \times B]$ $A \times [B] \rightarrow [A \times B]$	Every pair (a, b) with a from the first list and b from the second; cpl , cpr keep one side a single element.
11	$\text{cpp setify} = (\text{setify} \times \text{setify}) \frac{\exists \times \exists}{\exists}$ $\text{cpl setify} = (\text{setify} \times 1) \frac{\exists \times 1}{\exists}$ $\text{cpr setify} = (1 \times \text{setify}) \frac{1 \times \exists}{\exists}$	$[A] \times [B] \rightarrow P(A \times B)$	Forgetting the order with setify , cpp is the transpose of $\exists \times \exists$.
12	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	$F(PA) \rightarrow PFA$	$\text{cp}(F)$, the transpose of a relator's action on $\exists$; cpp implements it at $F(A) = A \times A$.
13	$\text{cp}(\text{list}) = \langle \frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} \rangle$ $\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{\exists \times \exists}{\exists}]$ $E(\text{cons})]$	$\text{list}(PA) \rightarrow P(\text{list}(A))$	$\text{cp}(\text{list})$: (5.12) $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$ at $\text{list}(\exists) = \langle [\text{nil}, (\exists \times 1) \text{ cons}] \rangle$, then the algebra expanded.
14	$\text{cplist} = \langle [\text{nil} \text{ wrap}, \text{cpp list}(\text{cons})] \rangle$	$[[A]] \rightarrow [[A]]$	$\text{cp}(\text{list})$ with every set a list: cpp in place of $\frac{\exists \times \exists}{\exists}$.

glue concat $\sqsubseteq (1 \times \text{concat}) \text{ cons}$

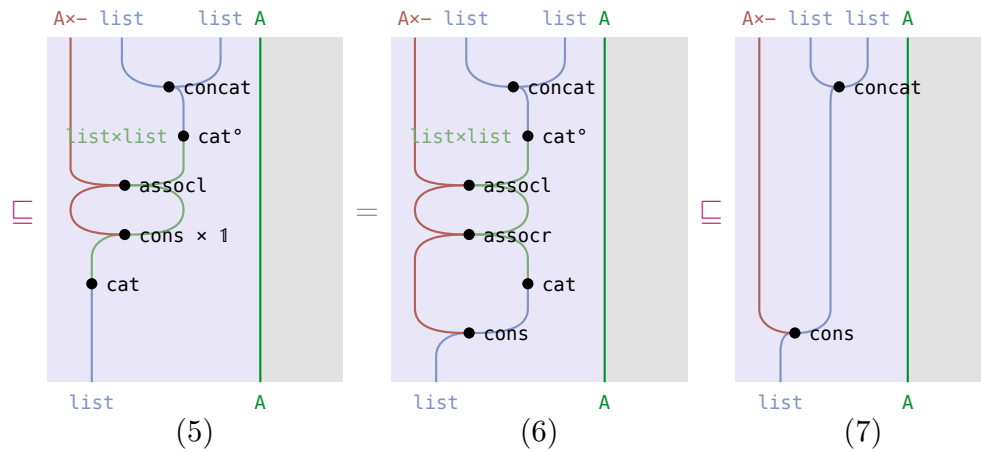
(5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(1×concat) cons**)

the starting composite



1	glue concat	definition of glue
2	$\triangleq (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cons } \text{concat}$	since $\text{concat} = ([\text{nil}, \text{cat}])$
3	$= (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 (1 \times \text{concat}) \text{ cat}$	naturality of assocl
4	$= (1 \times (\text{cons}^\circ (1 \times \text{concat}))) \text{ assocl } \text{cons} \times 1 \text{ cat}$	since $\text{concat} = ([\text{nil}, \text{cat}])$



5	$\sqsubseteq (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cat}$	since $(\text{cons} \times 1) \text{ cat} = \text{assocr } (1 \times \text{cat}) \text{ cons}$
6	$= (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{assocr } (1 \times \text{cat}) \text{ cons}$	since $\text{assocl } \text{assocr} = 1$ and $\text{cat}^\circ \text{ cat} \sqsubseteq 1$
7	$\sqsubseteq (1 \times \text{concat}) \text{ cons}$	

§5.6.1. $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$ — the set of subsequences is a fold of maps into sets

(5.6.1a)

$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$ — cup unions two sets.

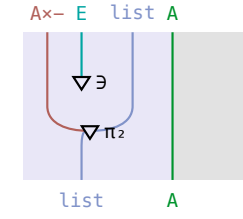
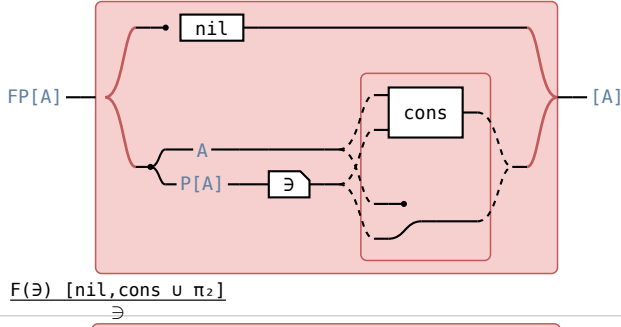
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : \text{PA} \times \text{PA} \rightarrow \text{PA}$

$$\frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}] \quad (5.6.1b)$$

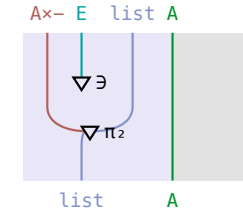
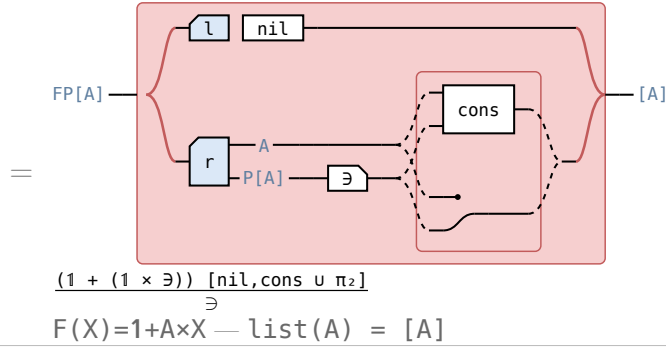
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$ at **subseq** = $\langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$, **subseq** = $\langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$.

circuit — the fork is $F([A])=1+A \times [A]: \text{nil}$ above, the pair below

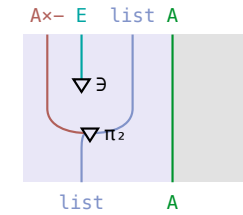
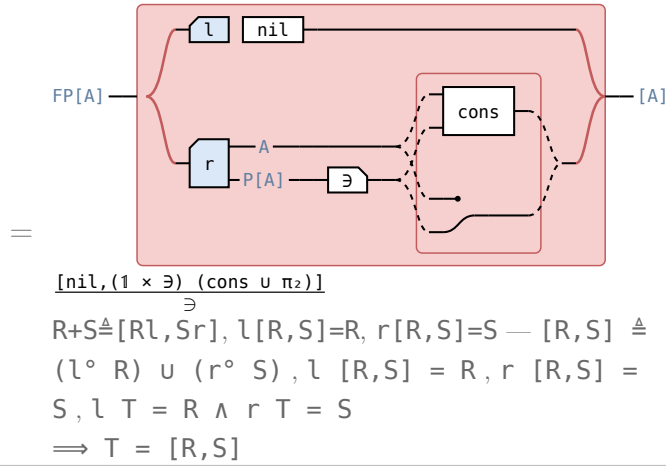
Hinze–Marsden



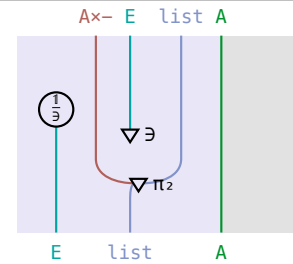
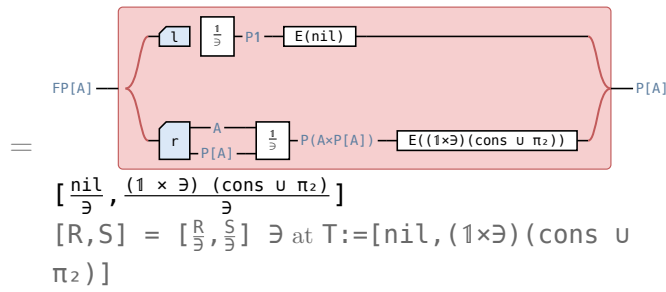
the π_2 operand of $\text{CONS} \cup \pi_2$ under the $1 \times \exists$ summand of $F(\exists)$, i.e. $(1 \times \exists) \pi_2$



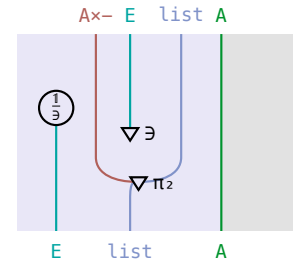
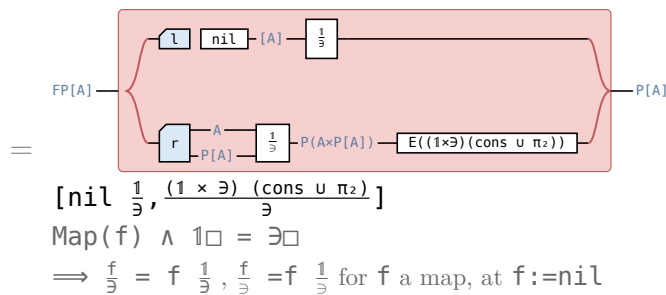
the same operand under $1+1 \times \exists$, whose $1 \times \exists$ summand it sits in



the π_2 operand of the second arm $(1 \times \exists) (\text{cons} \cup \pi_2)$

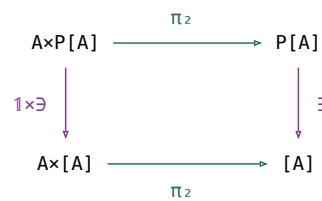


the π_2 operand under its $\frac{1}{\exists}$, the arm $(1 \times \exists) \pi_2 = \pi_2 \exists$ rewrites



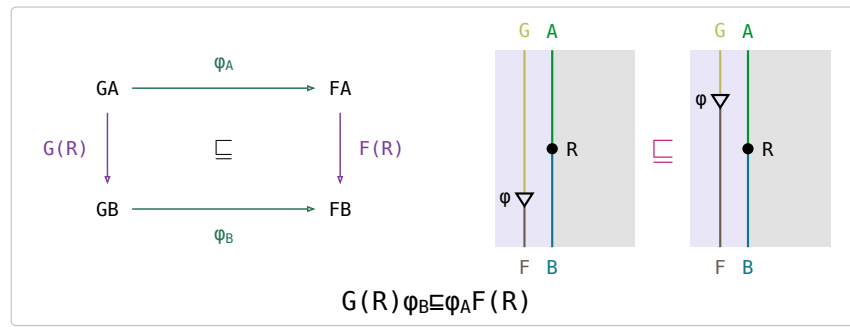
the same operand; the two rows differ only in the **nil** arm

(5.6.1c)



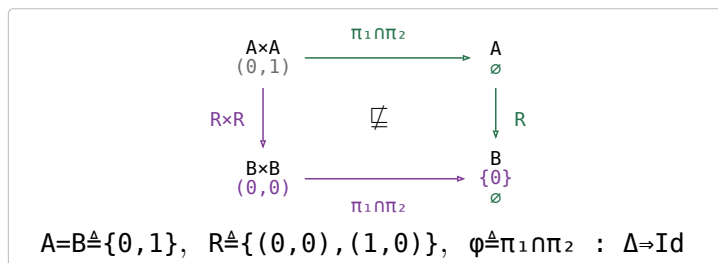
§5.7. Lax natural transformations (LaT)

$\varphi : G \Rightarrow F$ is **lax** at $R : A \rightarrow B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$; it is a **lax natural transformation** (LaT) when lax at every R . (5.7a)



(5.7c)

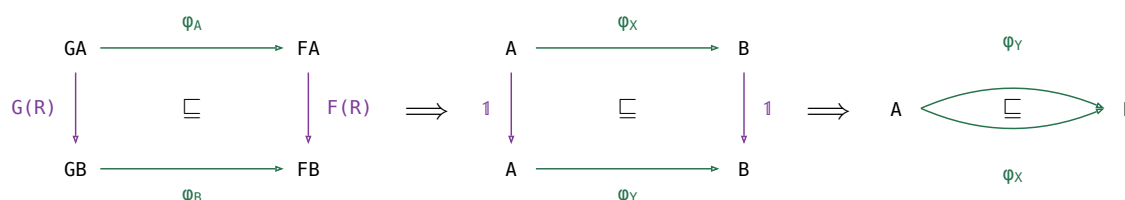
	closed under	commutative diagram	string diagram
1	composition $\psi\phi$	$H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\phi_B \sqsubseteq \phi_A F(R)$ give $H(R)(\psi_B \phi_B) \sqsubseteq (\psi_A \phi_A) F(R)$	
2	horizontal composition $\chi \circ \phi$	$\phi : G \Rightarrow F$ with $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and $\chi : L \Rightarrow K$ with $L, K : \mathcal{D} \rightarrow \mathcal{E}$ give $\chi \circ \phi : L \circ G \Rightarrow K \circ F$, the family $A \mapsto \chi_{GA} K(\phi_A)$ $L(G(R)) \chi_{GB} \sqsubseteq \chi_{GA} K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))$ is K applied to ϕ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\phi) : K \circ G \Rightarrow K \circ F$, K composed on the outside; $\phi := \mathbb{1}_G$ gives $\chi G : L \circ G \Rightarrow K \circ G$, G composed on the inside two candidates, $\chi_{GA} K(\phi_A)$ and $L(\phi_A) \chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
3	union $\phi \cup \psi$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A) F(R)$	
4	a relator K $K(\phi)$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ gives $K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))$	
5	product $\phi \times \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi \times \psi : G \times G' \Rightarrow F \times F'$ $(R \times S)(U \times V) = (RU) \times (SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle \phi, \psi \rangle = \triangleleft (\phi \times \psi)$, and its ONE cost is the lax copy law $R \triangleleft \sqsubseteq \triangleleft (R \times R) \rightarrow R \triangleleft \sqsubseteq \triangleleft (R \otimes R), R \multimap \sqsubseteq \multimap$	
6	coproduct $\phi + \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi + \psi : G + G' \Rightarrow F + F'$ $(R + S)(U + V) = (RU) + (SV)$ and monotonicity in both slots; the co-fork is the derived case $[\phi, \psi] = (\phi + \psi) \nabla$, and ∇ costs nothing the step it would need is $(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)$, the wrong direction of R ($S \cap T \sqsubseteq (R S) \cap (R T)$) a counterexample: $aa = (\emptyset, 1) \wedge bb = (\emptyset, 0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{\emptyset\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1)$ $\Rightarrow \neg((R \times R)(\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2) R)$	
7	meet — fails $\phi \cap \psi$		
8	Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$, quantified over R ; at $X' = X$ with R_A, R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference		

$$(5.7.1a)$$


(5.7.2a)

		lower — inside ONE allegory	upper — between allegories
1	0-cell	an object A	an allegory $\mathcal{C}$ — a whole lower tower
2	1-cell	a relation $R : A \rightarrow B$	a relator $F : \mathcal{C} \rightarrow \mathcal{D}$
3	2-cell	$R \sqsubseteq S$	a LaT $\varphi : G \Rightarrow F \triangleq \text{lax}$
4	order	$R \sqsubseteq S$	$\varphi \sqsubseteq \psi$ componentwise
5	union	$R \cup S$	$\varphi \cup \psi$, survives $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$
6	product	$R \times S \triangleq \times$	$\varphi \times \psi$, survives ; a TENSOR, not a categorical product; SYMMETRIC $\langle R, S \rangle$ $\pi_1 = \text{dom}(S) \circ R, \langle R, S \rangle \circ \pi_2 = \text{dom}(R) \circ S$ $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow (G(R) \times G'(R)) (\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A) (F(R) \times F'(R))$
7	coproduct	$R + S$	$\varphi + \psi$, survives ; a BIPRODUCT $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow (G(R) + G'(R)) (\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A) (F(R) + F'(R))$
8	meet	$R \sqcap S$	$\varphi \sqcap \psi$ componentwise, fails $aa = (\emptyset, 1) \wedge bb = (\emptyset, \emptyset) \wedge (R \times R)(aa, bb) \wedge b_1 = \{\emptyset\} \wedge b_1 = E(\pi_1 \sqcap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \sqcap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1) \Rightarrow \neg((R \times R) (\pi_1 \sqcap \pi_2) \sqsubseteq (\pi_1 \sqcap \pi_2) R)$
9	converse	R°	φ° , fails , oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$
10	zero object	z with $1 \ z = \emptyset$	the constant relator at such a z , initial and terminal

(5.7.3a)



with $\mathcal{C}$ two objects and one non-identity arrow $X \rightarrow Y$, a LaT $\text{const } A \Rightarrow \text{const } B$ is the pair $\varphi_X, \varphi_Y : A \rightarrow B$ with $\varphi_Y \sqsubseteq \varphi_X$ — the datum $R \sqsubseteq S$

not every LaT is constant: $\exists : P \Rightarrow Id \quad X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ \quad \pi_2 : (A \times -) \Rightarrow Id \quad (1 \times \exists) \circ \pi_2 = \pi_2 \circ \exists$